

prevent incidents

work safely

Safety is fundamental to everything we do.

reduce risk

Safety Aspects

Safety Aspects

- The safety-related signs used throughout this learning material:



DANGER

DANGER indicates a hazard with a high level of risk which, if not avoided, may result in serious injury or death.



WARNING

WARNING indicates a hazard with a medium level of risk which, if not avoided, could result in serious injury or death, or damage to the equipment.



CAUTION

CAUTION indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury, or damage to the equipment.

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Safety - Introduction

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- **Hazard prevention is the key to electrical safety.**
- Always consider electrical safety and use good preventive practices before performing any work, maintenance, or repair procedures.
- Always obey the applicable OSHA, EU-OSHA, governmental and corporate safety practices.
- Some of UPS's conditions that can affect safety are highlighted here.
- The practices and recommendations mentioned here are general and remind of the hazards of repairing and maintaining large power distribution devices. They do not supersede the abovementioned regulations.

	Wear eye protection
	General symbol for mandatory action
	Read the manual or instructions
	Disconnect from power source

Safety - Tools

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- When you do service calls or procedures on Eaton electrical equipment, obey the following rules concerning tools, testers, solvents, adhesives, and lubricants:
 - Make sure that electrical hand tools, such as power drills, are inspected regularly.
 - Replace worn and broken tools and test equipment with new ones.
 - Only use tools designed for the purpose.
 - Use insulated tools when there is any risk of electrical hazard or working close to electrized parts.



Safety - General

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- Obey the following safety rules for working with electrical and mechanical equipment in the maintenance and repair procedures:
 - The UPS must be used as intended. Obey the instructions given in the User's and Installation Guide.
 - Make sure that the installation and use of the UPS comply with all national and local safety regulations and procedures.
 - Make sure that all primary power switches installed downstream from the UPS are labelled as follows: "ISOLATE UPS (Uninterrupted Power Supply) before working on this circuit."



Safety - Environment

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- Obey the following rules:
 - Always inform the appropriate supervisor or manager of conditions or voltages that may threaten safety. Take all necessary steps to maximize safety.
 - Always look out for possible hazards, such as moist floors, non-grounded extension cables, power supplies, and missing safety grounds.
 - Do not make unauthorized changes or modifications to the equipment. This practice creates a hazard and unsafe equipment.
 - Before you start the equipment, ensure that other service and customer personnel are not exposed to unsafe conditions.
 - Do not wear loose clothing that can be trapped in the moving parts of a machine. Make sure that your sleeves are fastened.
 - If you are wearing a necktie or scarf, remove it before performing maintenance.
 - Tie up or otherwise secure long hair to make it safe.
 - To prevent strain on your back, lift the equipment or its parts by pushing up with your legs. Do not lift any heavy equipment or parts that cannot be lifted comfortably.
NOTE: Note the local lifting restrictions. Use a lifter whenever possible and applicable.
 - To prevent tripping hazards, always keep tool kits away from walk areas. If possible, keep the tool kit on or under a table.
 - Keep the unit area clean when you do maintenance and after completing the job.
 - While you service the unit, store removed unit covers in a safe place. Reinstall the covers before you return the unit to the customer.
 - Reinstall all safety devices, such as guards, shields, and ground wires, when maintenance is complete. Replace safety devices that are



DANGER

Do not work alone in potentially hazardous conditions or near equipment that has potentially dangerous voltages.

The risk of serious or fatal injury exists when retrieving dropped objects, such as screws, washers, and so on, from the inside of the unit. If tools or hardware are dropped in the unit, shut down the system before retrieving them. Use extreme caution if power is still present in the unit.

Safety - Electrical

Obey the following rules when you work on electrical machinery:

- To prevent loss of power to the customer's load, obey the shutdown and maintenance bypass procedures as instructed in the User's and Installation Guide. If recommended in this manual, switch the input and logic power off:
 - ✓ Before you replace the main components of the equipment
 - ✓ Before you work near power supplies
 - ✓ Before you inspect power supplies
 - ✓ Before you install changes in the machine circuits
 - ✓ Unless the maintenance documents specifically instruct otherwise, do not make unauthorized changes or modifications to the equipment. This creates a hazard and unsafe equipment.



WARNING

Do not service input, output, or bypass circuit breakers, transformers, or current transformers (CTs) with the power on.

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Safety - Electrical

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- If working on equipment that has exposed live electric circuits, obey the following precautions:
 - ✓ Wear proper personal protective equipment (PPE).
 - ✓ Make sure that another person who is properly trained in the power-off controls is always within a close distance to switch off the power if needed.
 - ✓ Do not wear jewellery, chains, metal frame eyeglasses, or other metal objects.
 - ✓ Only use insulated probe tips or extenders.
 - ✓ Use only one hand while you work on or near energized equipment. Keep the other hand in your pocket or behind your back to prevent electric current flow across the heart.
 - ✓ Do not touch objects that are grounded, such as metal floor strips, machine frames, or other conductors. Use suitable rubber mats. Obtain the mats locally, if necessary.
 - ✓ When using test equipment, set the controls as instructed in the equipment's operation or service manual. Use only properly insulated probes.



WARNING

Do not touch the unit and the oscilloscope at the same time. Use an adapter to isolate the oscilloscope from the earth ground. There can be a high voltage potential between the UPS unit frame and the oscilloscope.

Safety - Electrical

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- When you work with machines with voltages larger than 30 Vac or 42.4 Vdc, obey the special safety instructions in the UPS Service Manual and field service bulletins.
- The output neutral of the system is connected with the required neutral for the customer's load and should never be bonded to ground at the module output.
- Make sure that all battery breakers, including those on External Battery Modules (EBMs), are open before checking voltages.
- Do not use any conductive equipment like mirrors to inspect the unit inside.
- In case of an electrical accident:
 - Remove all power from the system.
 - Instruct another person to get medical aid.



WARNING

Never assume that power has been removed from a circuit. Use a voltmeter to check and confirm that power has been removed from the circuit.

Safety - Eyes



CAUTION

Wear safety glasses at all times.

Do not wear soft contact lenses when working on or around electrical equipment.

- Use additional caution:
 - when you use a hammer
 - when you use a power drill
 - when you solder parts
 - when you cut wire or remove steel bands
 - when you use solvents, chemicals, or cleaners
 - when you work in any other condition that might injure the eyes

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Safety – Maintenance/Service



DANGER

This UPS contains lethal voltages.

All repairs and service must be done by authorized service personnel only.

There are no user-serviceable parts inside the unit.

- Always wear appropriate eye protection.
- Remove restrictive or loose clothing and all jewellery and other conductive substances.
- Use the correct documentation and appropriate tools as outlined in the Service Manual.
- Use a static-secured work area and ESD procedures when performing component replacement or modifications.
- Make sure that power is disconnected before installing or servicing when possible.
- Obey all the CAUTION, WARNING, and DANGER notices on the equipment's inside and/or outside and contained in the Service Manual.
- Always comply with the more detailed safety precautions described in the appropriate sections in this Service Manual.
- Only Eaton-trained and certified personnel are authorized to do any service work related to UPS devices.
- **There are no user/serviceable parts inside the UPS**

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Safety – Maintenance/Service



CAUTION

Only qualified service personnel knowledgeable of batteries and the required precautions are allowed to do the installation or service work on batteries. Keep unauthorized personnel away from the batteries. Before you install or replace batteries, consider all the warnings, cautions, and notes concerning appropriate handling. Do not disconnect the batteries when the UPS is in Battery mode.

Make sure that your replacement batteries are of the same number and type as the battery that was originally installed in the UPS. See more accurate instructions on the UPS.

Before you connect or disconnect battery terminals, disconnect the charging source by opening the corresponding battery circuit breaker.

If the battery is inadvertently grounded, remove the source of the ground. All batteries in all UPS models are ungrounded.

Discard batteries according to your local disposal requirements. Do not dispose of batteries in a fire. When exposed to flame, batteries may explode.

To ensure proper cooling airflow and to protect personnel from dangerous voltages inside the unit, keep the UPS door closed and the front panels installed.

Do not install or operate the UPS system close to gas or electric heat sources. Keep the operating environment within the parameters stated in this document.



CAUTION

Keep the surroundings of the UPS uncluttered, clean, and free from excess moisture.

Obey all DANGER, CAUTION, and WARNING notices affixed to the equipment.



CAUTION

To prevent damage to the wiring channel and the wiring configuration in the UPS cabinet base when lifting or moving the cabinet, use only the left or right forklift slot to lift and move a packed cabinet.

Make sure that the forklift fork is inserted into the forklift slot at the horizontal position. Do not tilt the fork upwards.

Insert the forks all the way through the base.

Failure to comply with these instructions may cause damage to the wiring channels and the wiring configuration.

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Safety – Batteries

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WARNING

It is critical to properly set up the UPS for the proper battery configuration. Failing to do so can result in a thermal event of the batteries.



WARNING

Before you connect or disconnect batteries, make sure that the battery breaker at the front of the electronic module or EBM is in the OFF position.



WARNING

Make sure that the wires from the electronics module are connected to the positive (+), negative (-), and neutral (N) terminals of the battery trays. There is a risk of explosion if a battery or batteries are replaced by an incorrect type.



WARNING

Dangerous voltage is always present at the battery terminals. Batteries can present a risk of electrical shock or burn from high short-circuit current and high voltage. Obey correct precautions. Incorrect connection of batteries may cause electrical shock, fire, injury, or death.



CAUTION

When you remove or install batteries, use a battery lift, wear proper PPE, and obey proper lifting techniques.

Safety – Batteries



WARNING

The UPS is powered by its own energy source (batteries). The output terminals could be energized even when the UPS is disconnected from an AC source.

The UPS module interiors are unsafe until the DC power source is disconnected and the electrolytic capacitors are discharged. After disconnecting the utility power and the DC power, authorized service personnel must wait at least 5 minutes for capacitor bleed-off before attempting internal access to the UPS module.

To reduce the risk of fire or electric shock, install the UPS in a temperature and humidity controlled, indoor environment that is free of conductive contaminants. See temperature and humidity limits in chapter *Technical data* in this manual.

The ambient temperature limit must not be exceeded. Do not operate the UPS near water or excessive humidity. The system is not intended for outdoor use.

Before you start any installation or service work, make sure that all AC and DC power sources are disconnected. Power can come from multiple sources. Also ensure system grounding / PE continuity.

In a parallel system, the output terminals could be energized even when the UPS is turned off.



WARNING

Batteries present a risk of electrical shock or burn from high short-circuit current. Obey proper precautions.

Electric energy hazard. Do not attempt to alter any battery wiring or connectors. Attempting to alter wiring can cause injury.

Do not open or mutilate batteries. Released electrolyte may be toxic and is harmful to the skin and eyes.

Batteries can contain HIGH VOLTAGES, and CORROSIVE, TOXIC and EXPLOSIVE substances. Because of the battery string the output terminals can carry high voltage even when the AC supply is not connected to the UPS. Read the shutdown instructions carefully.

IMPORTANT: The battery may consist of multiple parallel strings. Make sure that you disconnect all strings before installation.

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- Follow the local safety instructions for UPS maintenance and the safety instructions from the User installation and Service manuals. **Refer to the manual for all safety aspects.**
- Review and refer to all safety warnings and cautions in the manual before installation.
- Build a clear, permanent, restricted access area around the system.
- **Only authorized, properly trained electrical operators should be able to access the system.**
- **Only authorized Eaton Customer Service Engineers or other qualified service personnel authorized by Eaton can service and maintain the system.**
- **There are no user/serviceable parts inside the UPS.**

	NOTICE
	This product shall be installed in a restricted access area where only the qualified personnel who are trained and have the knowledge of the product and the related safety precautions of the installation manual. "Restricted access area" is area accessible only to the electrically skilled persons and electrically instructed persons with the proper authorization.



DANGER
This UPS carries lethal voltages. All repairs and service must be done by authorized personnel only. There are no user-serviceable parts inside the UPS.

Eaton 93T 15-80 kW UPS

93T 15-80kW

Generations of experience

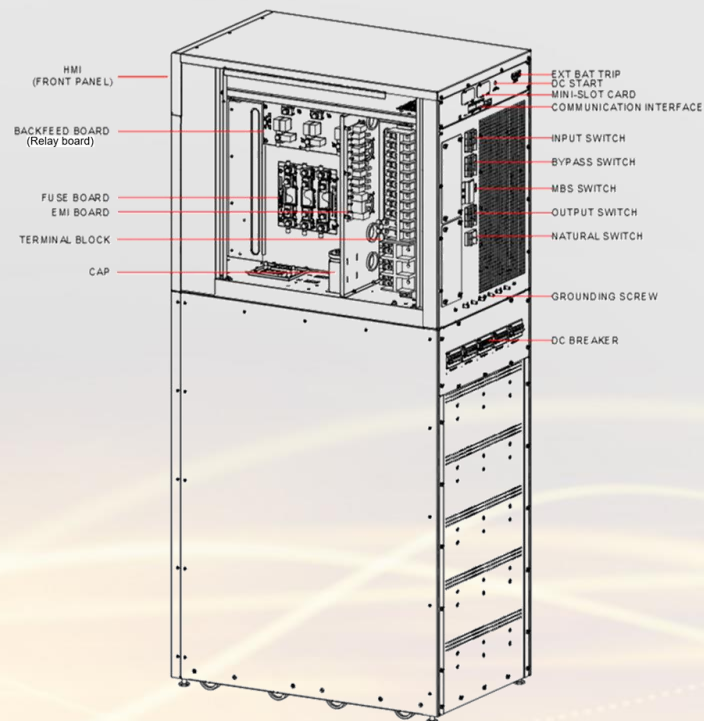
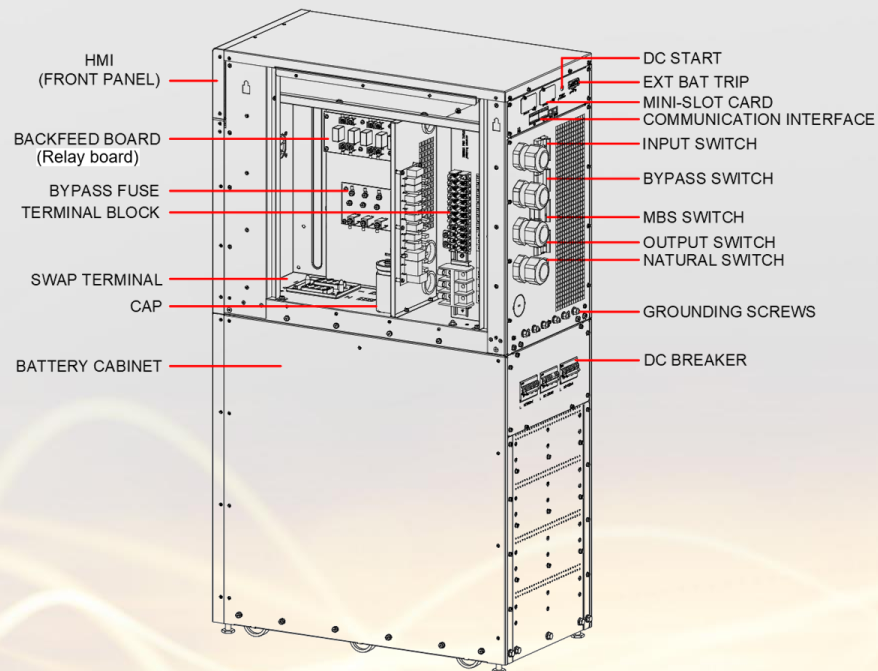
Over 60 years of UPS leadership



Eaton 93T 15-80 kW UPS

EATON 93T UPS

UPS Structural Overview



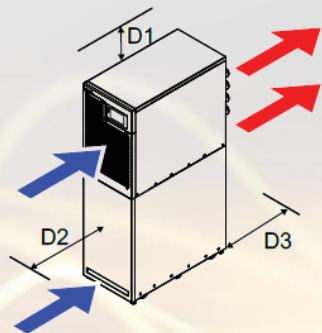
UPS Clearance

Table 6. UPS cabinet minimum clearances

Location		Clearance [mm]	
		15–40 kW	60–80 kW
D1	Above the cabinet	300	300
D2	In front of the cabinet	900	900
D3	Behind the cabinet	300	400

If needed, allow more clearance for wiring.

Figure 12. 93T UPS cabinet clearances



Commissioning



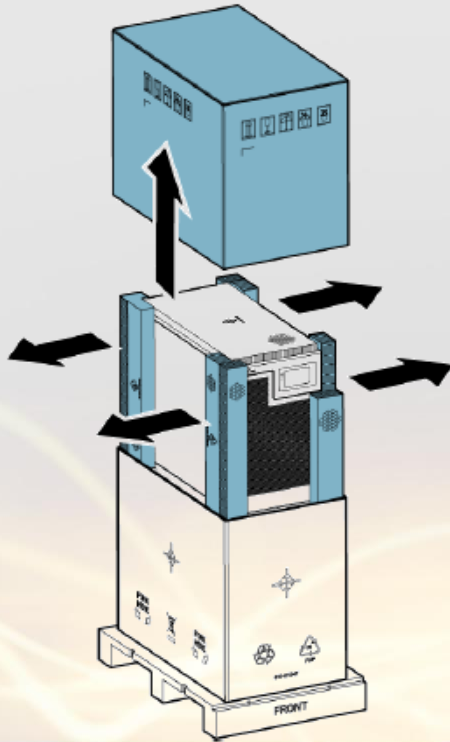
Mechanical inspection



Mechanical inspection during commissioning phase

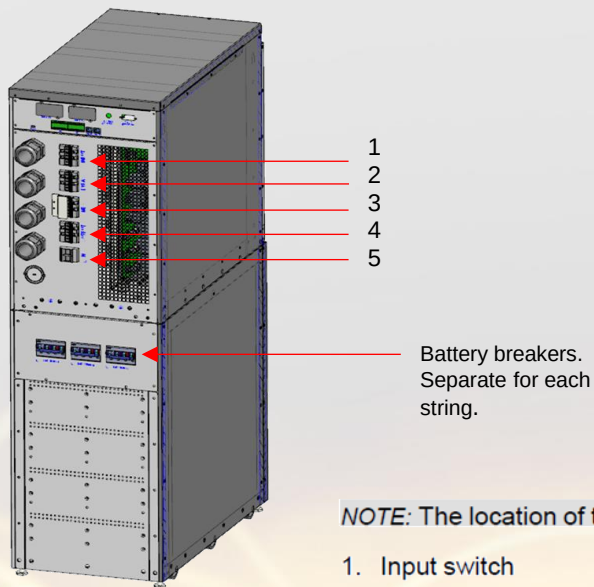
- ✓ Package check
- ✓ Visual inspection
- ✓ Electrical inspection
- ✓ Metering without power applied
- ✓ Short circuit check

Unpacking and installation



- Follow instructions from the User and Installation manual

Unpacking and installation



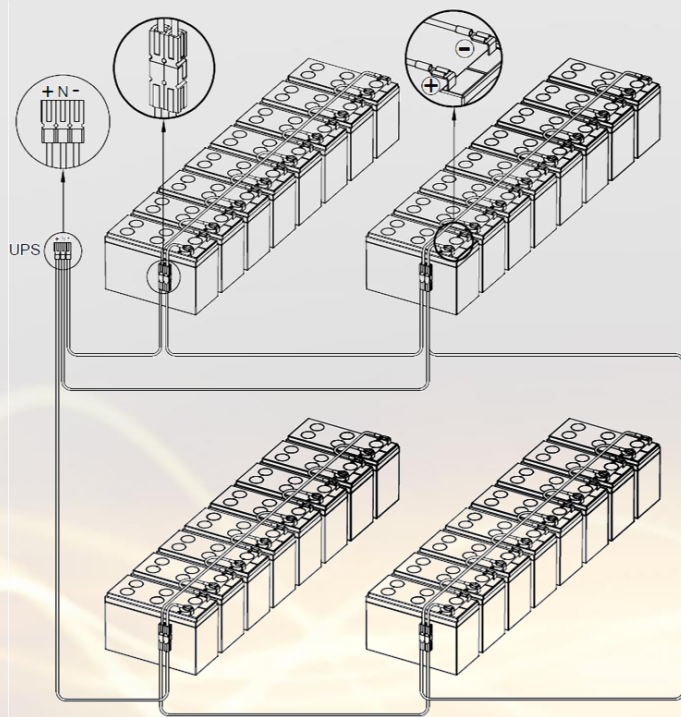
Battery breakers.
Separate for each
string.

NOTE: The location of the switches is the same in all 93T series UPSs.

- | | |
|------------------------------------|-------------------|
| 1. Input switch | 4. Output switch |
| 2. Bypass switch | 5. Neutral switch |
| 3. Maintenance bypass switch (MBS) | |

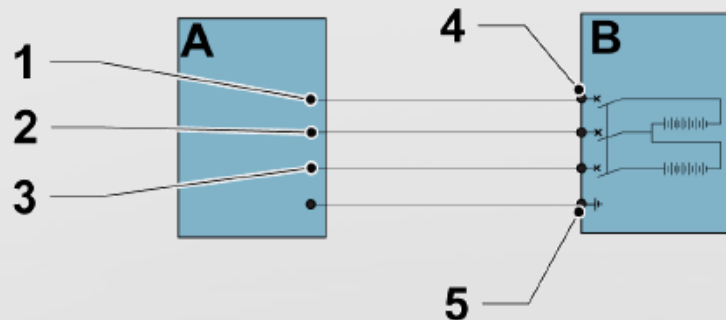
- After the cables are installed, keep all feed breakers open, including the input breaker, bypass breaker, and neutral and battery breakers.

Unpacking and installation



- If the unit contains internal batteries, use a known-good DMM to make sure that power is not present in the cable terminals.
- There is a separate battery breaker for each string in the internal battery cabinet.

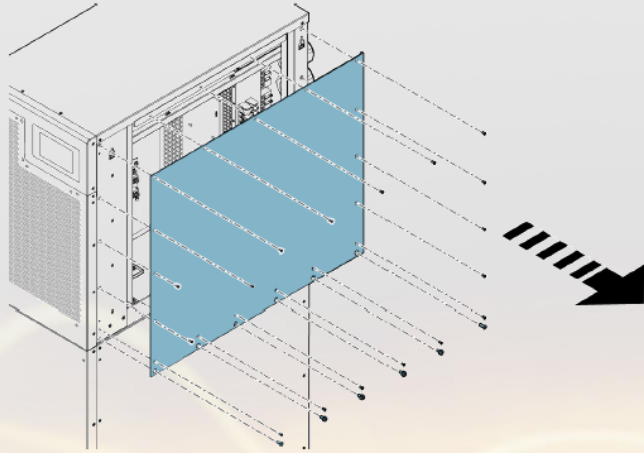
External batteries



- In case of installing external batteries please contact an Eaton service representative.

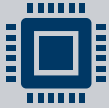
- | | | | |
|----|--------------------------|---|------------------|
| 1. | Battery + | A | UPS |
| 2. | Neutral | B | External battery |
| 3. | Battery - | | |
| 4. | Battery breaker | | |
| 5. | Protective earthing (PE) | | |

Unpacking and installation



- Remove the side cover

Unpacking and installation



Measure and ensure there is no short circuit in the input connection terminals.

Keep the input breaker open.



Measure and ensure there is no short circuit in the output connection terminal.

Electrical inspection and configuration



Electrical inspection

Make sure that the UPS input and bypass breakers (in case of dual input) are open.

Close the feed breaker(s) to apply utility and bypass power to the unit. (external breakers from facility)

Measure the voltage at input connection terminals:

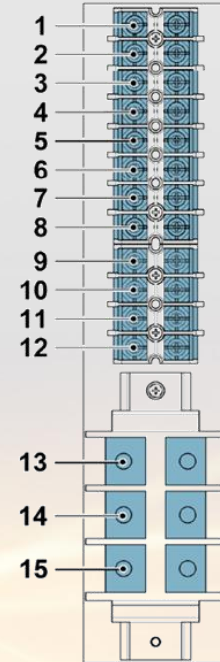
Use a phase rotation meter (or similar) to make sure that the phase rotation is correct.

Close the UPS input and bypass switches. Close all battery breakers.

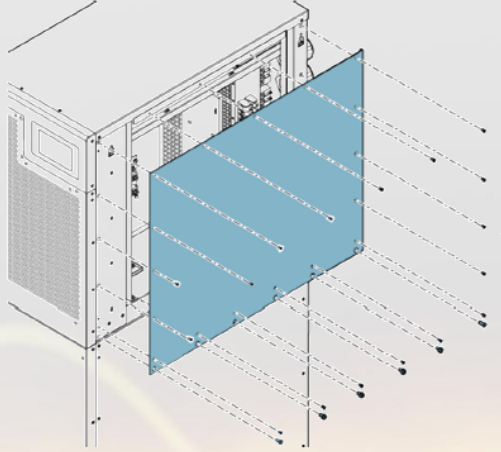
Measure the voltages phase-to-phase, phase-to-neutral, and phase-to-ground.

Make sure that the voltages are within the acceptable limits.

Make sure that there is less than 5 VAC from neutral-to-ground.

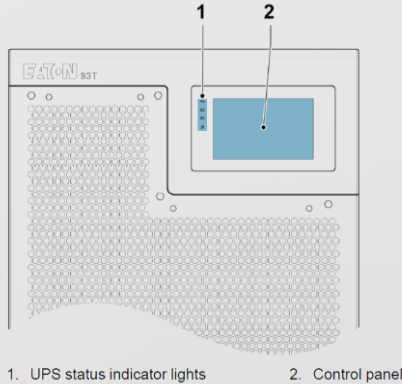


Electrical inspection



- Re-install the side cover

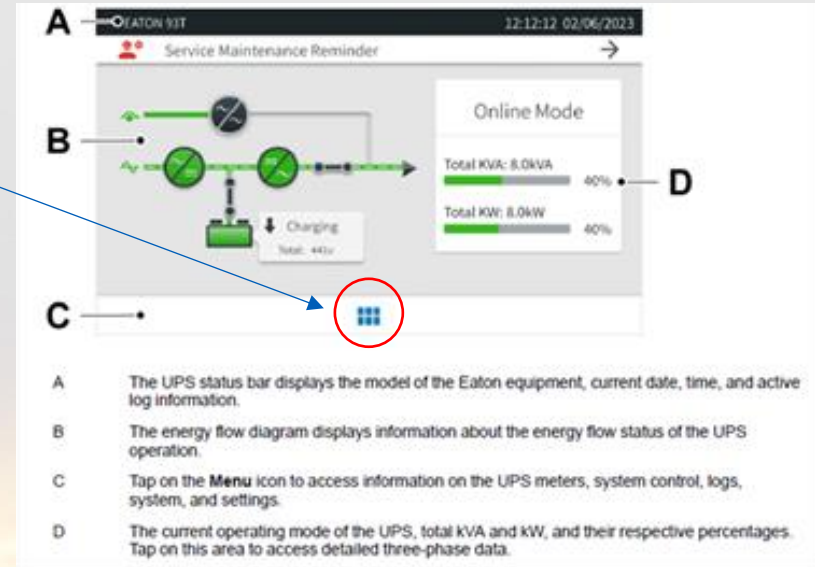
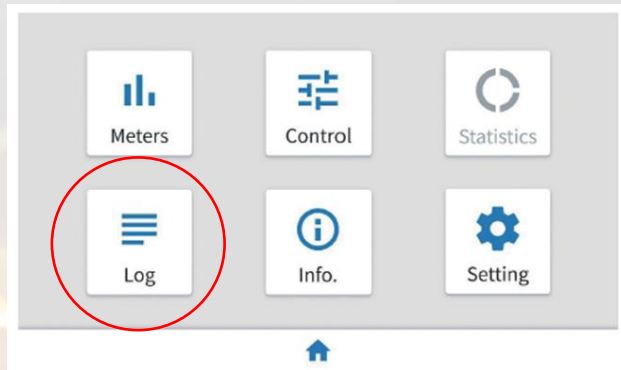
Electrical inspection



- Ensure the HMI display energizes
- The following alarms/events should be active:
 - Battery switchgear open
 - Configuration error

Electrical inspection

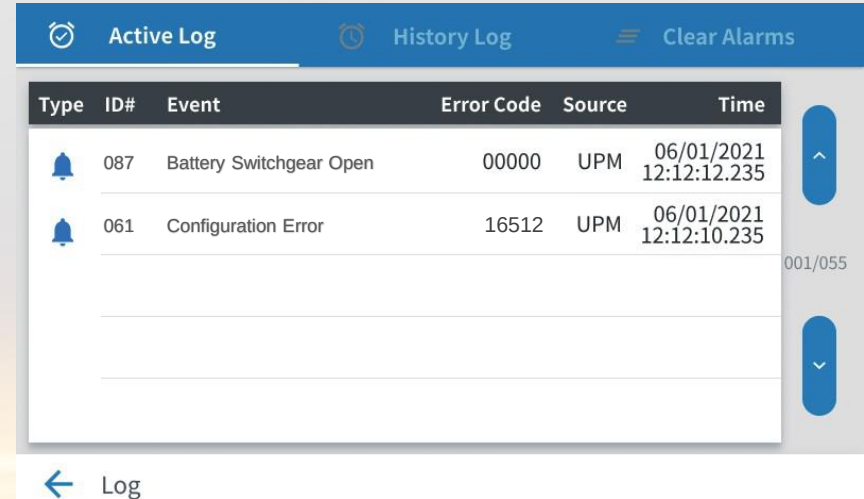
- Click the MENU [⊗] button from the display to access the menu screen
- Select the Log button



Active events – Log screen

- Active events after power is applied to UPS
- Reasons:
 - Battery switchgear open → relays will close automatically when the unit rectifier starts up

LOG Screen:



The screenshot shows a mobile application interface for monitoring UPS events. At the top, there is a blue header bar with three tabs: 'Active Log' (selected), 'History Log', and 'Clear Alarms'. Below the header is a table with the following columns: Type, ID#, Event, Error Code, Source, and Time. The table contains two entries, each preceded by a blue bell icon. To the right of the table is a vertical scrollbar with up and down arrows. At the bottom of the screen is a white bar with a blue back arrow and the text 'Log'.

Type	ID#	Event	Error Code	Source	Time
	087	Battery Switchgear Open	00000	UPM	06/01/2021 12:12:12.235
	061	Configuration Error	16512	UPM	06/01/2021 12:12:10.235

001/055

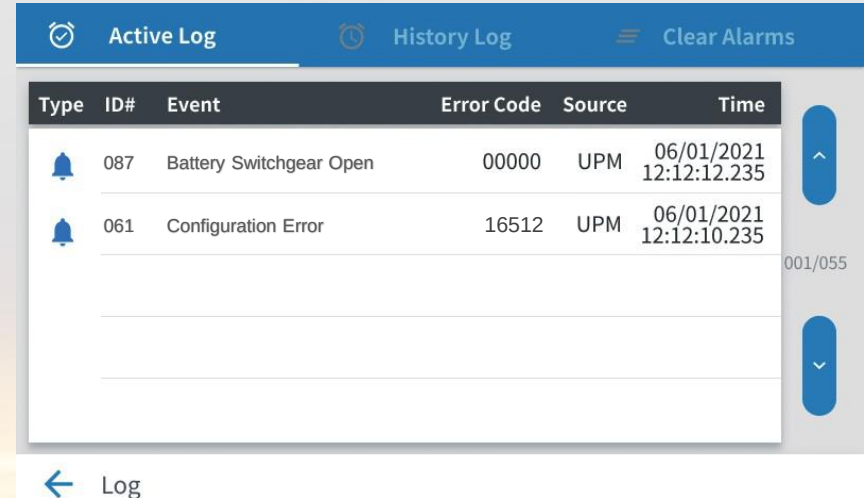
← Log

Active events – configuration error

Error code 16512.

- Battery cells per string need to be configured.
 - See the next slide
- Setup required. Needs to be cleared.
 - See the next slide

LOG Screen:

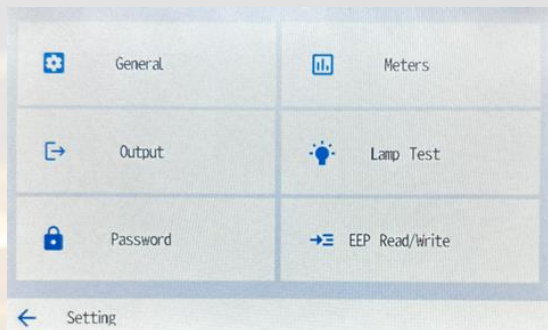
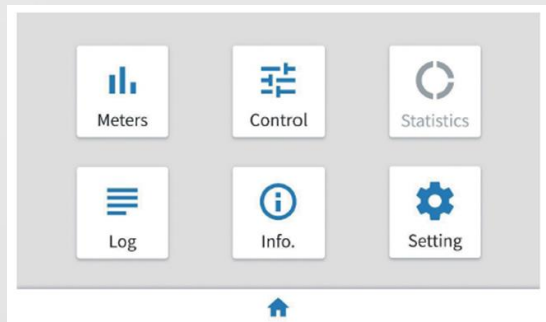


Type	ID#	Event	Error Code	Source	Time
	087	Battery Switchgear Open	00000	UPM	06/01/2021 12:12:12.235
	061	Configuration Error	16512	UPM	06/01/2021 12:12:10.235

001/055

← Log

Display



- Go to Settings
 - Use the password from the registration email.
- Go to EEP Read/Write

Display

- Battery cells per string
 - Type Address 05302
 - Enter value based on unit type
 - use the highlighted value from the table
 - Press *Change*
 - Press ← to go back to the Settings screen

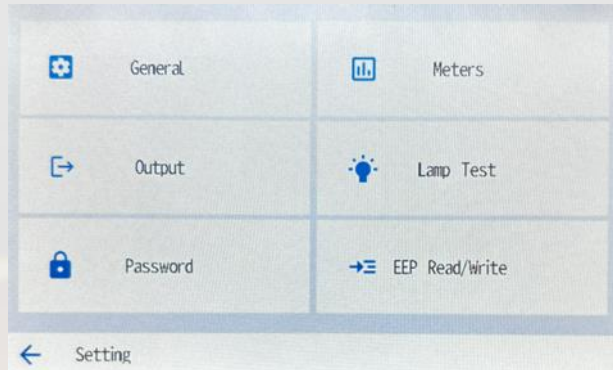
Reference	Description	Amount of batteries per string	Amount of cells per string
93T15KMBSBI	93T 15kVA IB 1x36x9Ah	36	00216
93T20KMBSBI	93T 20kVA IB 2x32x7Ah	32	00192
93T30KMBSBI	93T 30kVA IB 2x36x9Ah	36	00216
93T40KMBSBI	93T 40kVA IB 3x32x9Ah	32	00192
93T60KMBSBI	93T 60kVA IB 4x40x9Ah	40	00240
93T80KMBSBI	93T 80kVA IB 5x40x9Ah	40	00240

The screenshot shows a device display with the following elements:

- Address:** A field with five empty boxes for input.
- Value:** A field with five empty boxes for input.
- Buttons:** A blue "Read" button and a purple "Change" button.
- Numeric Keypad:** A 3x3 grid of buttons containing numbers 1 through 9, a back arrow with an 'x', 0, and an "OK" button.
- Status Bar:** At the bottom, it shows a left arrow, "EEP", and "Read/Write".

Display

- Go to EEP Read/Write



Display – Required setup

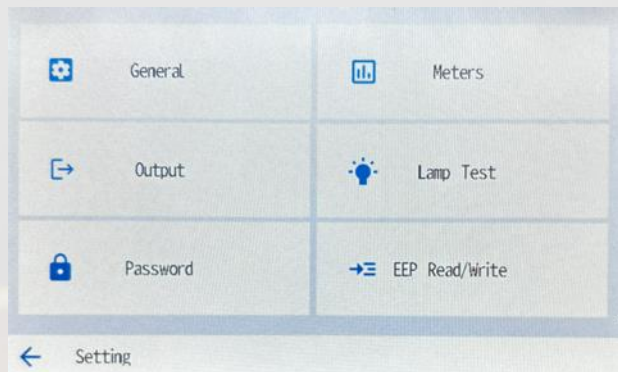
- Type Address 05069
- Enter value 00000
- Press *Change*
- Press ← to go back to the Settings screen

The screenshot shows a device display with the following elements:

- Address:** A label followed by a 5-digit numeric input field.
- Value:** A label followed by a 5-digit numeric input field.
- Buttons:** A blue "Read" button and a purple "Change" button.
- Keypad:** A 3x3 grid of buttons containing digits 1-9, 0, a back arrow with an 'x', and an "OK" button.
- Footer:** A blue arrow pointing left followed by the text "EEP Read/write".

Display

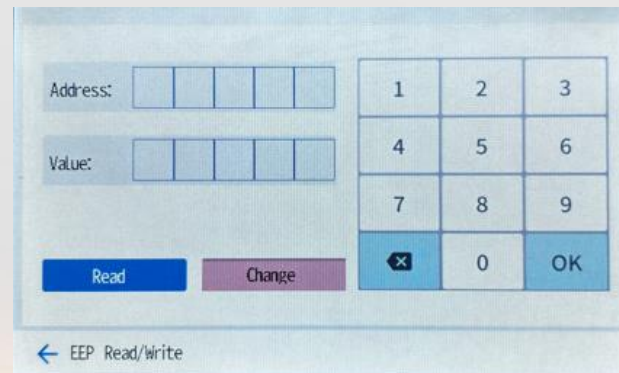
- Go to EEP Read/Write



Display – Optional setup

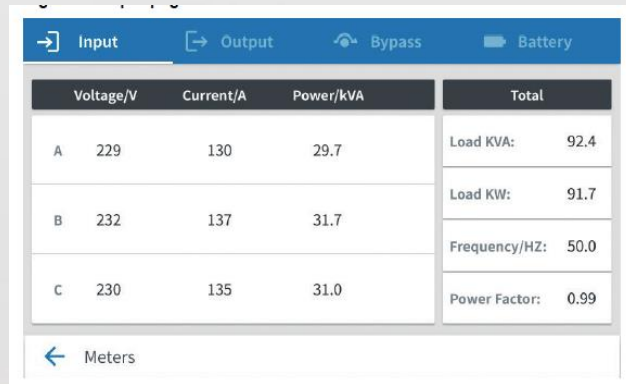
If remote EPO is needed, do the following setup:

- Type Address 05067
- Enter value 00001
- Press *Change*
- Press ← to go back to the Settings screen
- Press ← to go back to the main menu screen.
- See the User's and Installation manual for EPO wiring instructions.



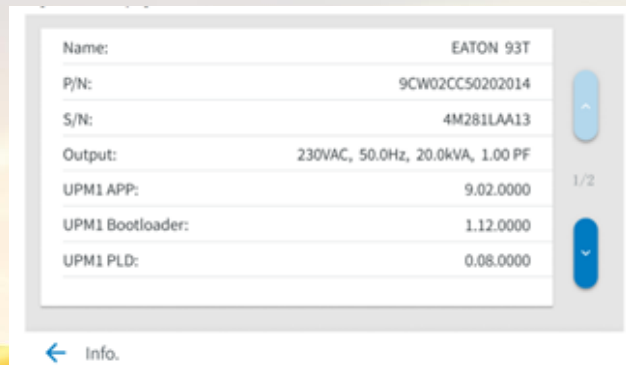
Electrical inspection

- Examine the HMI display and complete the following steps:
 - Ensure the input, bypass, and battery meters are correct: Menu → Meters
 - To examine the firmware versions: Menu→Info
 - To examine the part number (CTO) and serial number: Menu→Info
 - To configure the date, time, and language settings: Menu→Setting→General
 - Press  to go back to mimic screen



	Voltage/V	Current/A	Power/kVA	Total
A	229	130	29.7	Load KVA: 92.4
B	232	137	31.7	Load KW: 91.7
C	230	135	31.0	Frequency/HZ: 50.0
				Power Factor: 0.99

← Meters



Name:	EATON 93T
P/N:	9CW02CC50202014
S/N:	4M281LAA13
Output:	230VAC, 50.0Hz, 20.0kVA, 1.00 PF
UPM1 APP:	9.02.0000
UPM1 Bootloader:	1.12.0000
UPM1 PLD:	0.08.0000

← Info.

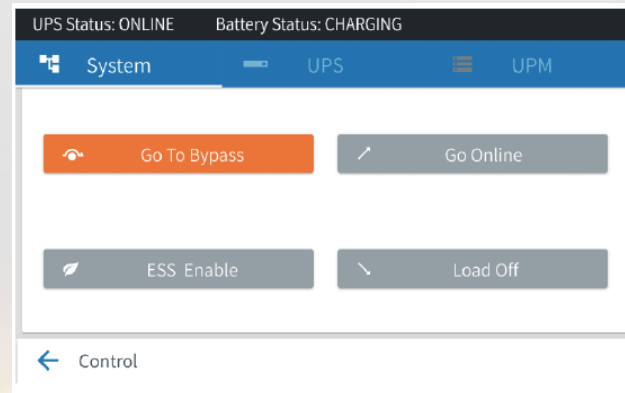
Operational inspection



Starting the UPS

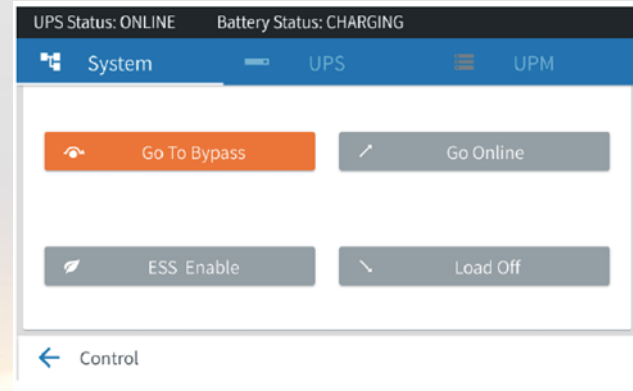
Follow these steps:

1. From the HMI display, select **Go To Bypass**.
2. Examine the UPS meters and status. Make sure that the UPS operates correctly on bypass.
3. From the HMI display, select **Load off**.



Starting the UPS

4. From the HMI display, select *Go Online*.
 - Examine the UPS meters and status. Make sure that the UPS operates correctly in **Online mode**.
 - Open the input breaker. Make sure that the unit operates correctly in **Battery mode**.
 - Close the input breaker. Make sure that the unit transfers back to **Online mode**.
 - Transfer the UPS to **Bypass mode** by selecting *Go To Bypass*. Make sure that the unit operates correctly in Bypass mode.
5. On the HMI display, select *Go Online*.





Powering Business Worldwide

Eaton.com